

SEMESTER-VIII

UME842: MECHANICS OF COMPOSITE MATERIALS				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: The objective for this course is to develop an understanding of the elastic analysis of composite materials. This course also introduces the concept of unidirectional composites, short fiber composites, orthotropic lamina, laminated plates and beams.				
Syllabus Introduction: Definition, characteristics, classification, fabrication of composites, fiber-reinforced composites, applications of composites. Properties of Unidirectional Composites: Longitudinal behavior of unidirectional composites, initial stiffness, load sharing, longitudinal strength and stiffness, transverse stiffness and strength, prediction of shear modulus, prediction of Poisson's ratio, failure modes. Short-Fiber Composites: Introduction, theories of stress transfer, approximate analysis of stress transfer, average fiber stress, modulus and strength of short-fiber composites. Analysis of an Orthotropic Lamina: Introduction, orthotropic materials, stress-strain relations and engineering constants, Hooke's law and stiffness and compliance matrices, general anisotropic material, compliance tensor and compliance matrix, maximum-stress theory, maximum-strain theory, maximum-work theory. Analysis of Laminated Composites: Introduction, laminate strains, variation of stresses in a laminate, resultant forces and moments: synthesis of stiffness matrix, symmetric laminates, unidirectional, cross-ply, and angle-ply laminates, determination of laminae stresses and strains. Analysis of Laminated Plates and Beams: Introduction, governing equations for plates, equilibrium equations, equilibrium equations in terms of displacements, application of plate theory, bending, Buckling, analysis of laminated beams, governing equations for laminated beams. Research Assignment: Research assignment will constitute collection of literature, problem formulation (mathematical model) required for design consideration and experimental characterization (mechanical testing), environmental issues, metal and ceramic matrix composites, nanocomposites, bio- composites etc. The students work in groups to test samples of composite materials, scan for defects, SEM study etc. This assignment also includes technical report writing and seminar presentation.				
Course Learning Outcomes (CLOs) The students will be able to: 1. determine the properties of fiber and matrix of composite material in different orientations. 2. predict the elastic properties of both long and short fiber composites. 3. relate stress, strain and stiffness tensors using ideas from matrix algebra.				

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4. analyze a laminated plate in bending, including finding laminate properties from lamina properties.
5. determine the failure strength of a laminated composite plate.

Text Books

1. Agrawal, B.D. and Broutman, L. J., Analysis and Performance of Fiber Composites, John Wiley & Sons, New York (1990).
2. Daniel, I.M. and Ishai, O., Engineering Mechanics of Composite Materials, 2nd edition, Oxford University Press, (2005).

Reference Books

1. Mallick, P.K., Fiber-Reinforced Composites: Materials, Manufacturing, and Design, Second edition, CRC Press (1993).
2. Herakovich, C.T., Mechanics of Fibrous Composites, John Wiley & Sons, New York (1998).
3. Christensen, R.M., Mechanics of Composite Materials, John Wiley & Sons, New York (1979).

Evaluation Scheme:

S N	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessionals (Assignments/Projects/Tutorials/Quizzes)	25

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